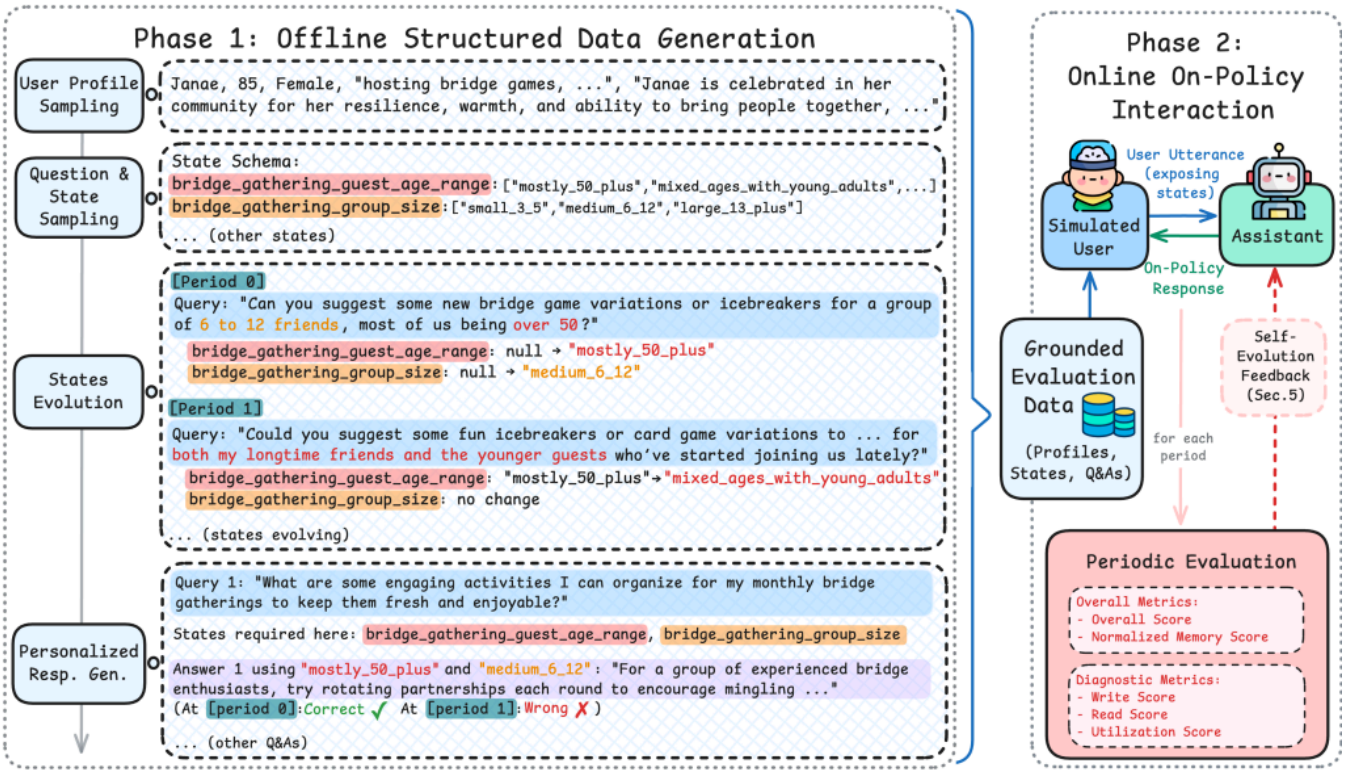
AMEMGYM: INTERACTIVE MEMORY BENCHMARKING FOR ASSISTANTS IN LONG-HORIZON CONVERSATIONS

> #Benchmark #Memory #Dataset



- What problems does this framework solve?
  - Traditionally, we use an offline strategy to evaluate the LLM memory, i.e., we generate a set of traces and send the traces to the benchmark.
  - This can cause some problems, such as it cannot reflect how the following conversation affects the inference of the assistant.
  - This can lead to mislead in further optimization (the result can be not persuasive)
- What does this framework do?
  - Instead of using an offline strategy, we provide the assistant with an environment to interact with a "user" in an online manner.
  - GENERATING STRUCTURED DATA FOR ON-POLICY INTERACTION
    - In this step, we create the dataset.
    - Firstly, sample some personas.
    - Then, for a persona, we sample some questions. For each question, extract state-related *variables* needed along with all possible *domains* that are necessary to answer the question.

      All of such variables and domains will be added together to be a global schema.
    - Each user is tracked through a timeline, and each state on the timeline is represented by a

vector σ. The framework will generate some event $e$that pushes the change of state: $\sigma_{-1} \rightarrow \sigma$. The states will be implicitly exposed by User Utterance.
        - Generate personalized answers for each predefined question as the ground truth.
    - ON-POLICY INTERACTION
        - Using utterance as a start, the benchmark posts a question to the assistant.
        - Based on the utterance-contained question, the assistant answers.
        - Now it's the turn of the User LLM to shine. It generates responses based on user profiles, the current actual state (σ_t), and the latest conversation context. It no longer follows a script but instead “improvizes” based on the assistant's recent performance.
    - It not only measures the overall quality of the answer (normalized: random guess as lower bound and cheating as upper bound) but also feedbacks the kind of mistakes (State Utilization/Read/Write Failure).
- Experiment and Observations
    - On-Policy Evaluation Matters: Offline assessments tend to overestimate certain models' capabilities, particularly for Agentic Memory systems where offline evaluations fail to capture the relative strengths of read and write strategies during dynamic interactions.
    - Long-Term Memory is the Weak Link: As dialogues lengthen, native LLMs experience a steep decline in performance, whereas purpose-built memory agents like Agentic Write demonstrate superior results.
    - Self-Evolution Capability: This is the most intriguing aspect. Because AMEMGYM provides environmental feedback, agents can autonomously evolve better memory strategies by experimenting, receiving feedback (e.g., identifying where they made errors), and then leveraging LLMs to automatically modify their memory prompts.
- My Thought
    - This can be a good baseline of self-representative agent (a copy of you) benchmark, as this framework provides a solid evaluation of memory functionality. However, we also need other evaluation perspectives, such as whether tone matches the actual tone.
    - Tone match can be hard to solve, as it is highly subjective. Perhaps Contrastive Learning?
- Work Cited
    - Paper
    - Chat History